

Business Sales Performance Analytics

Superstore Sales Dashboard & Insights Report

Data Science & Analytics Internship — Task 1 Submission

Dataset: Superstore Sales Dataset (Kaggle) | Tool: Microsoft Excel

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Executive Summary

Over the four-year period covered by this dataset (2014–2017), the business generated total sales of \$2,297,200.86 and total profit of \$286,397.02, an overall profit margin of 12% across 5,009 orders. Revenue shows a clear upward trend over time, with recurring seasonal peaks. However, revenue and profit do not move together throughout the business: several high-revenue products, sub-categories, and regions are quietly losing money, driven almost entirely by excessive discounting.

The single clearest pattern in the data is the relationship between discount level and profit: every discount band above 30% operated at a net loss, while orders sold at 0–10% discount generated more profit on their own than the company's total profit across all orders combined. This, along with underperformance in the Furniture category and the Central region, forms the basis of the recommendations in this report.

About This Analysis

Objective: to analyze business sales data and answer the core questions a real business owner would ask — which products generate the most revenue, how sales change over time, which categories or regions are most profitable, and where the business should focus to grow faster.

- Dataset: Superstore Sales Dataset (Kaggle) — 9,994 transaction-level rows covering Sales, Profit, Discount, Region, Category, Sub-Category, and Product detail from 2014–2017.
- Tool used: Microsoft Excel — PivotTables, PivotCharts, Slicers, and an interactive KPI dashboard.
- Data cleaning performed: removed duplicate rows; identified and corrected a mixed-format date field (some dates stored as real Excel dates, others as US-format text) using a custom MONTH/DAY/YEAR conversion formula; standardized numeric data types for Sales, Profit, Discount, and Quantity; and added helper columns for Order Year, Order Month, and Profit Margin %.
- Analysis performed: revenue trend over time, top 10 products by revenue and profit, category/sub-category performance, region/state performance, and the impact of discount level on profit.

Key Performance Indicators

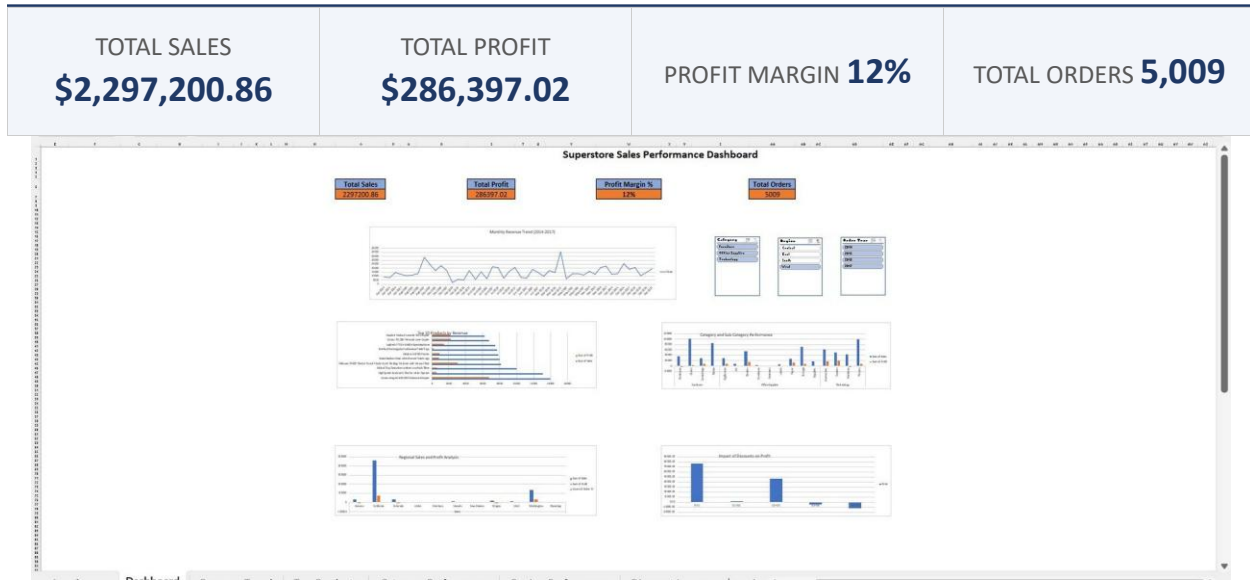


Figure 1. Full interactive dashboard with KPI cards, region/category/year slicers, and all five core charts.

Revenue Trends Over Time

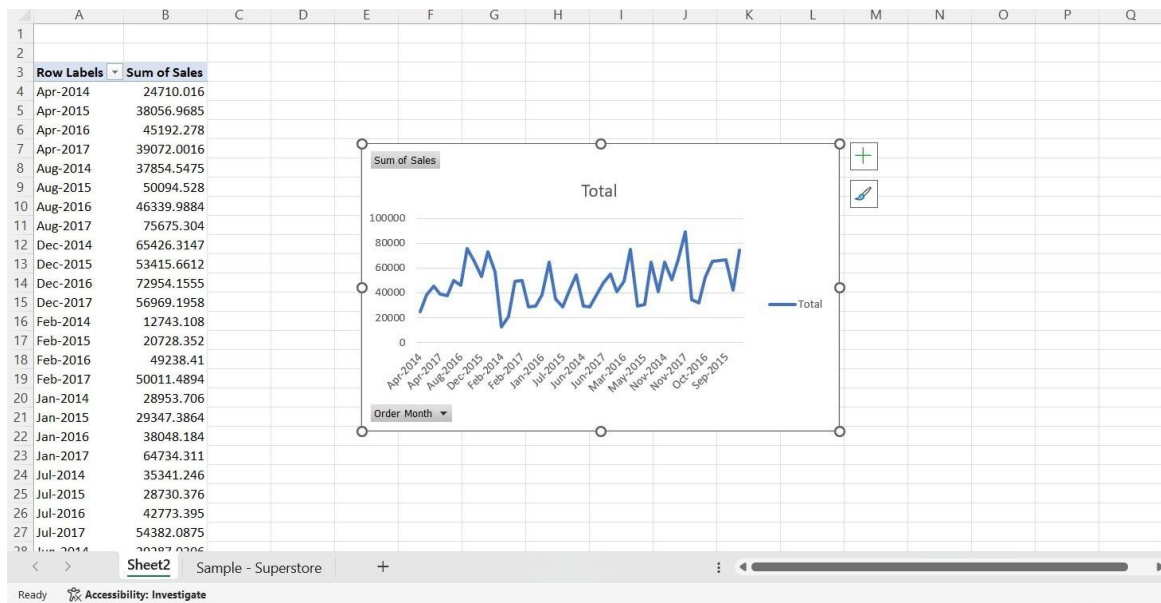


Figure 2. Monthly revenue trend, Order Month vs. Sum of Sales (2014–2017).

Monthly revenue grew substantially over the four-year window — from a low of \$12,743.11 in February 2014 to a peak of \$75,675.30 in August 2017, the single highest-selling month in the dataset. The trend line shows recurring seasonal spikes, generally concentrated around August–September and November–December, consistent with back-to-business and year-end purchasing cycles. Recommendation: align inventory planning, staffing, and marketing spend with these recurring peak months rather than spreading budget evenly across the year.

Top-Selling Products

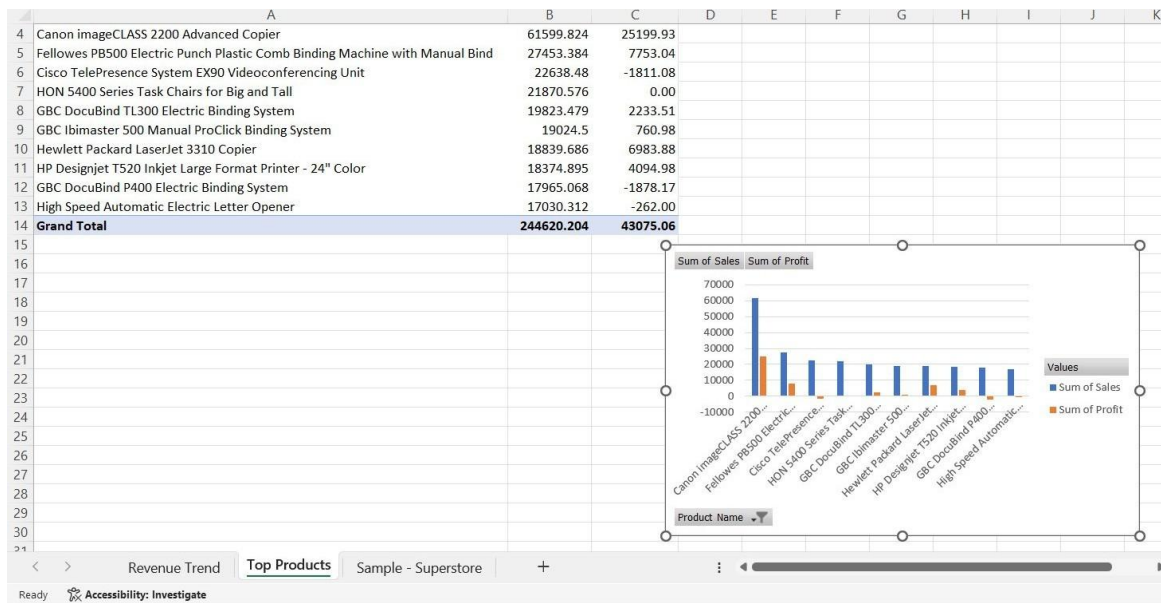


Figure 3. Top 10 products by revenue, with Sum of Sales and Sum of Profit.

The top 10 products by revenue together generated \$244,620.20 in sales (about 10.6% of total company revenue) but only \$43,075.06 in profit — a reminder that high sales volume does not automatically mean high profitability. Notably, three of these ten best-selling products are actually losing money: the Cisco TelePresence System EX90 (–\$1,811.08 profit on \$22,638.48 in sales), the GBC DocuBind P400 Electric Binding System (–\$1,878.17 profit), and the High Speed Automatic Electric Letter Opener (–\$262.00 profit). The HON 5400 Series Task Chairs sold \$21,870.58 worth of inventory at exactly \$0.00 profit. By contrast, the Canon imageCLASS 2200 Advanced Copier is the standout performer — \$61,599.82 in sales and \$25,199.93 in profit, a margin of roughly 41%, and a strong candidate for continued promotion.

Category & Sub-Category Performance

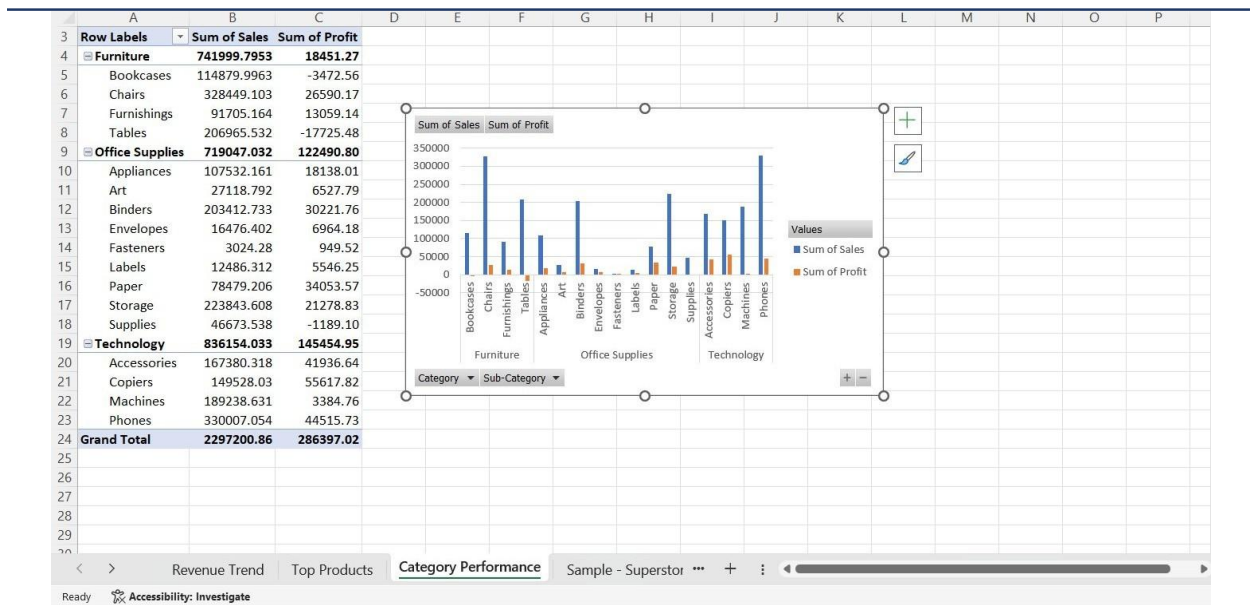


Figure 4. Sales and profit by Category, broken down by Sub-Category.

Category	Sales	Profit	Margin
Furniture	\$741,999.80	\$18,451.27	2.5%
Office Supplies	\$719,047.03	\$122,490.80	17.0%
Technology	\$836,154.03	\$145,454.95	17.4%

Furniture generates the second-highest revenue of the three categories (\$741,999.80) but by far the lowest margin, at roughly 2.5%, compared to about 17% for both Office Supplies and Technology. The cause sits at the sub-category level: Tables lose \$17,725.48 despite \$206,965.53 in sales, and Bookcases lose \$3,472.56 despite \$114,879.10 in sales — the only two sub-categories in the entire dataset operating at a net loss. Technology is the strongest category overall, led by Phones (\$330,007.05 sales / \$44,515.73 profit) and Copiers, which post the highest sub-category margin in the business at over 37% (\$149,528.03 sales / \$55,617.82 profit).

Regional Performance

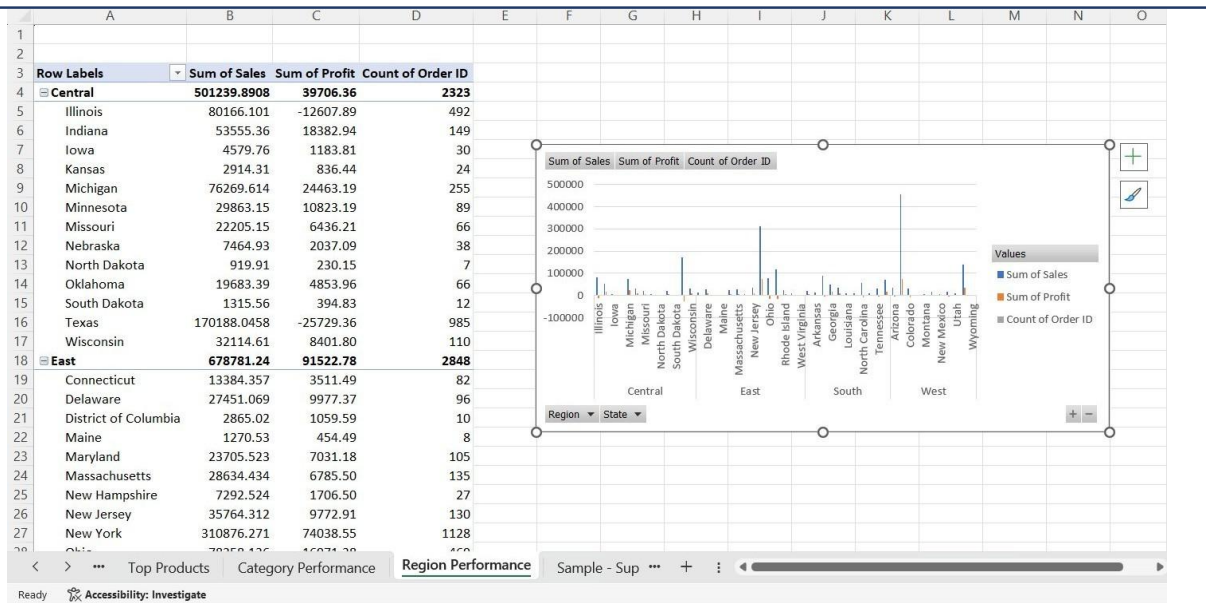


Figure 5. Sales, profit, and order count by Region, broken down by State.

East is the strongest region for profitability, with \$678,781.24 in sales and \$91,522.78 in profit (about 13.5% margin), while Central trails behind at \$501,239.89 in sales but only \$39,706.36 in profit (about 7.9% margin) — roughly half of East's margin despite still contributing meaningful revenue. Within Central, two states stand out as clear problem areas: Illinois posts a net loss of -\$12,607.89 on \$80,166.10 in sales, and Texas is the single largest loss-maker in the regional breakdown at -\$25,729.36 on \$170,188.05 in sales. These two states alone account for the bulk of Central's underperformance and are the natural starting point for a regional profitability review.

Discount Impact on Profit

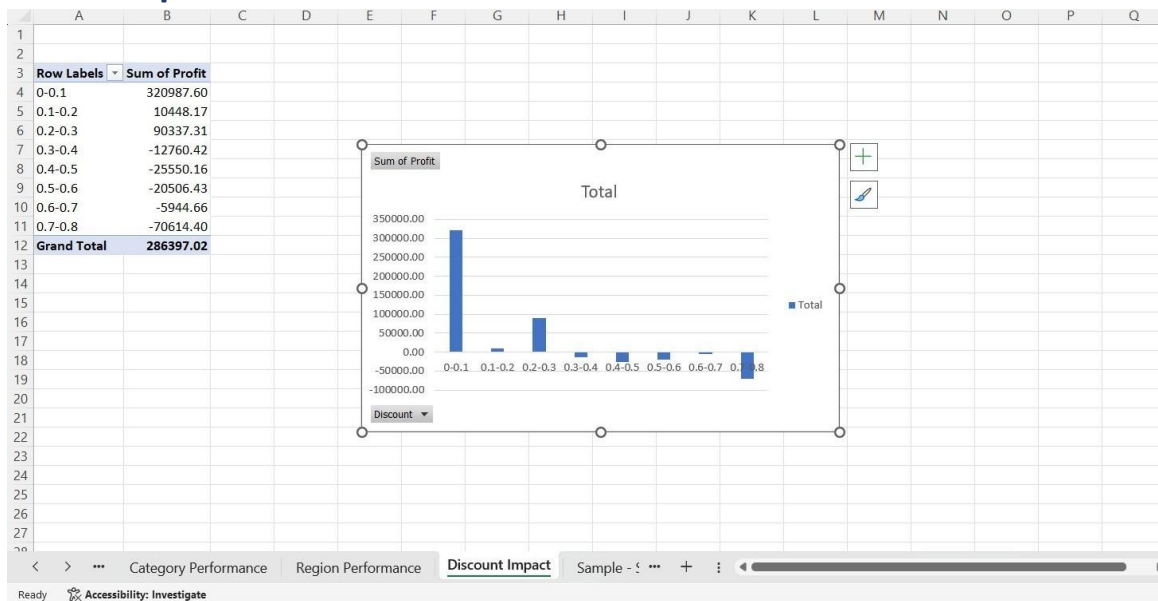


Figure 6. Total profit grouped by discount band (0–10% through 70–80%).

This is the most decisive pattern in the dataset. Orders sold at a 0–10% discount generated \$320,987.60

in profit on their own — more than the company's entire total profit of \$286,397.02, meaning every other discount band combined is a net drag on overall profitability. Profit turns negative once discounts exceed roughly 30%: the 30–40% band loses −\$12,760.42, the loss deepens through 40–50% (−\$25,550.16), and it bottoms out at −\$70,614.40 in the 70–80% band. There is a temporary rebound at 20–30% (+\$90,337.31), but this appears to be an isolated exception rather than a sustainable pattern — beyond 30% discount, the relationship with profit is consistently and significantly negative.

Key Insights & Recommendations

1. Cap discounts at 20% company-wide. Sales below a 10% discount drive almost all of the company's positive profit, and no band above 30% has produced a profitable outcome in four years of data. A hard discount ceiling at the point of sale is the single highest leverage, lowest-effort fix available to this business.
2. Re-price or renegotiate the Furniture category, specifically Tables and Bookcases. These are the only two sub-categories operating at a net loss across the entire dataset — either supplier costs need renegotiating, or discount permissions on these specific items need to be tightened before any further sales push.
3. Audit the Central region, especially Illinois and Texas. Both states are losing money despite meaningful sales volume; a targeted review of discounting practices, shipping costs, or return rates specific to these states is recommended.
4. Invest further in Technology, especially Phones and Copiers. These carry the strongest margins in the business and represent the clearest opportunity for profitable growth — incremental marketing spend here is likely to outperform the same spend on Furniture.
5. Re-evaluate three specific underperforming “top sellers”: the Cisco TelePresence System EX90, the GBC DocuBind P400, and the High Speed Automatic Electric Letter Opener. All three rank in the top 10 by revenue yet lose money — high sales volume is currently masking poor unit economics on these specific SKUs.
6. Plan around the Aug–Sep and Nov–Dec seasonal peaks identified in the revenue trend. Weighting inventory, staffing, and marketing budget toward these recurring high-demand months should improve return on spend versus a flat, year-round allocation.

Tools & Deliverables

- Analysis tool: Microsoft Excel (PivotTables, PivotCharts, Slicers, interactive KPI dashboard).
- Dataset: Superstore Sales Dataset, sourced from Kaggle.
- Deliverable 1: this report (insights and recommendations).
- Deliverable 2: the underlying Excel workbook, Superstore_Sales_Dashboard.xlsx, containing the cleaned dataset, five PivotTable/PivotChart sheets, and the interactive Dashboard sheet.

- Deliverable 3: a GitHub repository documenting the project (see README), with a link to the dashboard file or a screen recording/screenshot walkthrough.